An Intelligent IoT Architecture for Continuous Vehicle Health Monitoring and Early Fault Detection

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***Abstract*—Monitoring of key parameters of modern vehicles is essential for their safe operation, reliable functionality, and optimal performance. Unusual situations like overheating, high power consumption, and motor overload may cause equipment failure, inefficiency, and other faults. Conventional methods of monitoring do not include real-time monitoring, remote accessibility, and fault predictions. The present paper considers a Smart Vehicle Health Monitoring System based on the IoT and ML technologies. The system is designed on the basis of the ESP32 microcontroller, LM35 temperature sensor, ACS712 current sensor, BO motor, alerting buzzer, and Blynk cloud server for monitoring and fault detection. Data from sensors are collected in real-time mode by ESP32 microcontroller and analyzed in terms of classification of vehicle operation as NORMAL, WARNING, and CRITICAL modes. To evaluate the feasibility of data-driven classification, several machine learning algorithms including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and K-Nearest Neighbor were evaluated using experimentally collected temperature and current data. The machine learning analysis is intended as a comparison of classification techniques, whereas the deployed embedded system performs real-time fault detection using threshold-based logic. The processed data as well as system status can be transferred to the Blynk cloud platform via WiFi. The buzzer alerting system can be triggered under the abnormal operating conditions like overheating and overload of the vehicle. Thus, the developed system provides a low cost and intelligent solution for real-time diagnostics and maintenance of the vehicles.**

***Index Terms*—IoT, ESP32, Vehicle Health Monitoring, LM35 Sensor, ACS712 Current Sensor, Blynk Cloud, Embedded Systems, Real-Time Monitoring, Fault Detection, Predictive Maintenance, Smart Vehicles, Cloud Monitoring, Edge Computing, Automotive Diagnostics, Wireless Sensor Monitoring.**

I. Introduction

Present-day automobiles have developed into advanced embedded systems, which must be monitored to guarantee their safety, reliability, and effective operation. Such factors as temperature, current, voltage, and loads play an essential role in automotive performance. In case of overheating and overload, motors become damaged, suffer from electrical malfunctions, lose efficiency, and break down unexpectedly. Conventional monitoring systems are generally performed manually and do not provide timely diagnoses, remote access, or preventive maintenance. Recent researches on IoT technologies allow for effective monitoring and fault detection in automotive systems [1]-[4]. Modern automobiles include a variety of electronic subsystems that are responsible for ensuring their safe and effective operation. All of these subsystems interact constantly in order to achieve this goal through their embedded controllers and communication systems. Advanced technologies such as sensor installation and wireless communication have facilitated the diagnostic process and vehicle automation significantly. With growing system complexity, however, it becomes increasingly challenging to monitor the condition of the automobile and detect any malfunctions. Consequently, reliable and affordable monitoring is extremely relevant to modern automotive systems [5]-[6]. The advancements in the field of embedded systems and IoT technologies have transformed the vehicles into a cyber physical system. They can communicate and take intelligent decisions remotely. Scientists have conducted studies on integration of sensors, wireless modules, cloud computing and edge computing for diagnostics and predictive maintenance of vehicles [7]-[9]. Rajesh Kumar et al. developed an IoT-based vehicle monitoring system with the help of temperature and vibration sensors to achieve vehicle diagnostics and predictive maintenance [10]. It proved through their study that the monitoring via cloud makes the vehicle safe as well as efficient in its functioning. On the other hand, Patel et al. presented a wireless health monitoring system for automobiles by utilizing embedded sensors along with wireless technology [11]. Sharma et al. developed a predictive maintenance strategy for electric cars employing machine learning and cloud computing [12]. This was achieved through the use of sensor data for detecting abnormalities and predicting potential failure prior to any failure occurrence. In a similar case, Ramesh et al. developed an IoT based monitoring system for the monitoring of motors employing temperature and current sensors [13]. The monitoring was done to detect overload situations that might lead to motor failure. In recent IEEE journal articles focusing on smart transportation systems, researchers have investigated the role

of embedding sensors, edge computing, cloud dashboards, and wireless communication in vehicle diagnostics and predictive maintenance [14]-[15]. However, despite such improvements brought by existing systems, there are problems that need to be addressed, including costly implementation, inadequate edge computing capabilities, inability to classify faults in real time, and unavailability of affordable cloud-based visualization tools. For this reason, this paper suggests a Smart Vehicle Health Monitoring System consisting of ESP32, LM35 Temperature Sensor, ACS712 Current Sensor, BO Motor, Buzzer Alarm, and Blynk Cloud Dashboard. This solution monitors temperature and current values in real time, detects possible faults on the edge, identifies operational statuses of either NORMAL, WARNING, and CRITICAL types, and communicates data to a cloud dashboard through WiFi communication protocol.

II. Proposed System

The suggested Smart Vehicle Health Monitoring System aims at monitoring critical parameters of the vehicle such as temperature and current through the Internet of Things technology. The suggested system makes use of ESP32 as the processing and communication module, LM35 sensor to measure the temperature, ACS712 sensor to measure the motor current, BO motor is used as a laboratory-scale dynamic load for validating the proposed monitoring architecture. Although it does not represent a production automotive traction motor, it enables controlled evaluation of sensing, embedded processing, and cloud communication, buzzer as the fault notification module, and Blynk cloud-based platform to provide real-time monitoring. The LM35 sensor measures the temperature readings of the motor and electric component. On the other hand, the ACS712 sensor measures the motor current under varying loads. After the acquisition of the sensor values, they are further processed by the ESP32 microcontroller where edge-level processing takes place. Based on the acquired threshold conditions, the system determines its operational state. According to the obtained sensor values, the suggested system categorizes the working state of the vehicle into: NORMAL WARNING CRITICAL Under normal operation, the system keeps updating the sensor readings to the cloud dashboard by utilizing WiFi for communication. In situations where abnormal operation is encountered due to overheating or overload, the ESP32 triggers the buzzer alarm and updates the system status to the cloud dashboard. The design for the system entails the use of multi-layer IoT architecture, which comprises the sensing layer, edge processing layer, communication layer, cloud layer, and application layer. Under the multi-layer IoT architecture, the sensing layer is used to collect data from the sensors, edge processing layer is utilized for analyzing data at the edge through ESP32, while the communication layer uses WiFi for transferring data. The proposed system has many advantages, such as: Vehicle tracking in real-time, Visualization through cloud, Affordable cost

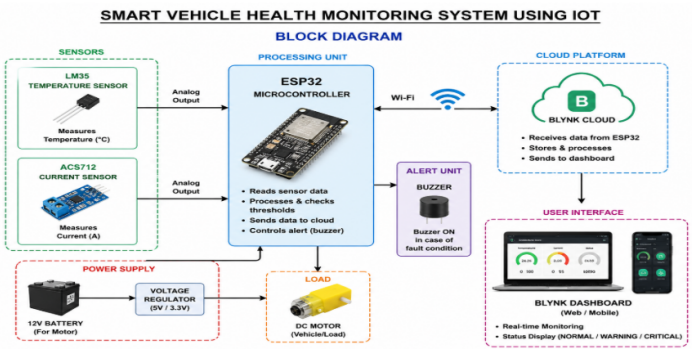


Fig. 1: Architecture of the proposed IoT-enabled smart vehicle health monitoring system

of setup, Fault detection at the edge, Proactive maintenance system, Scalable Internet of Things architecture and Web dashboard remote monitoring. In contrast with the existing vehicle tracking systems, the proposed vehicle tracking system conducts local fault analysis, then sends the results to the cloud, which helps decrease latency and increase system reliability. Temperature and current sensors provide more accurate fault detection and improve vehicle safety. As shown in Fig. 1.the proposed system, LM35 temperature sensor and ACS712 current sensor are connected to the ESP32 microcontroller. The ESP32 interacts with the Blynk cloud server through WiFi communications.

*1) **Machine Learning Integration**:* Besides the real-time monitoring, machine learning algorithms will be used in order to predict the health of the vehicle. Temperature and current measurements collected from multiple operating conditions were stored for analysis. Machine learning models were evaluated to compare their capability to classify experimentally observed operating conditions. In the deployed embedded system, real-time decisions continue to rely on threshold-based safety logic implemented on the ESP32, while machine learning analysis is used only for performance evaluation. Different machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and K-Nearest Neighbor are tested and the predictions are made based on sensor readings.

III. METHODOLOGY

The designed Smart Vehicle Health Monitoring System follows a comprehensive approach that includes sensing, edge computing, cloud communication, live monitoring, and fault diagnosis based on Internet of Things (IoT) architecture. This process includes various stages to facilitate real-time monitoring and effective vehicle diagnosis.

*1) **Data Acquisition**:* In the initial stage, live data from sensors is obtained using the LM35 temperature sensor and ACS712 current sensor. The LM35 sensor provides real-time temperature readings of the motor and electronics, while the

ACS712 sensor determines the current consumed by the motor under various load conditions.

*2) **Sensor Interfacing**:* Both of these sensors interface with the ESP32 microcontroller through the analog input port. The ESP32 microcontroller continuously captures analog input data from sensors and converts them to digital values.

*3) **Data Processing**:* The processing of the collected data involves converting them into the relevant parameters such as the temperature in degree Celsius and current in amperes.

*4) **Threshold-Based Fault Detection**:* Comparison is made between the processed sensor data and predefined threshold values for abnormal situations. Operating situation is categorized as follows: NORMAL WARNING CRITICAL In case of excessive value in temperature or current, then the situation will be identified as faulty situation.

*5) **Edge Level Analysis**:* Analysis and decisions are made by the ESP32 locally before transferring the data to the cloud platform. Edge level analysis cuts down communication delays and enhances response during critical faults.

*6) **IoT Platform**:* WiFi communication is used between ESP32 and the Blynk platform. Virtual pins are used for transferring sensor data like temperature and current data of the system to the cloud platform.

*7) **Cloud Monitoring and Visualization**:* The Blynk cloud platform monitors and visualizes sensor data and system status at a web dashboard. Vehicle health status monitoring is done by a remote user using the web dashboard.

*8) **Alert Mechanism**:* A buzzer alarm system is embedded to provide immediate alerts when fault occurs. When there is a case of an overload condition, the buzzer gets activated.

*9) **Machine Learning Based Classification**:* The dataset that has been collected from sensors is preprocessed for training various machine learning algorithms. Features like temperature and current readings are considered as input parameters for the algorithm.Machine learning models were trained using the experimentally collected dataset to compare classification performance. These models were evaluated independently of the threshold-based embedded decision logic.

Fig.2 compares the classification accuracy of five machine learning algorithms evaluated using the collected temperature and current sensor dataset. Among the tested models, Random Forest achieved the highest accuracy of 87.3%, followed by Decision Tree (84.7%) and Logistic Regression (80.05%). These results indicate that Random Forest provides the most reliable classification performance for comparative vehicle health assessment.

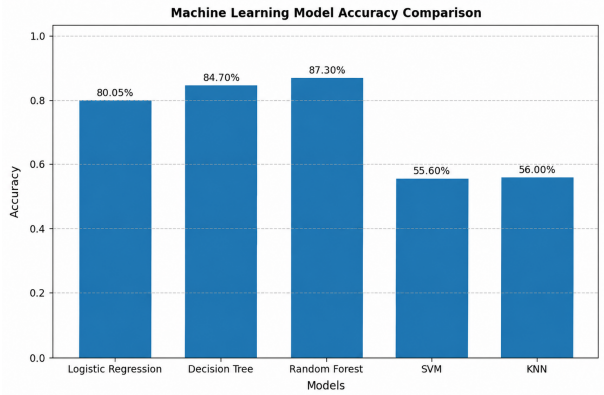


Fig. 2: Machine Learning Model Comparison Dashboard

A. SOFTWARE IMPLEMENTATION

The proposed Smart Vehicle Health Monitoring System software design is implemented using the Arduino Integrated Development Environment (IDE) with the programming language Embedded C/C++. The ESP32 microcontroller is used in this design as the main hardware. The software performs various tasks including sensor data collection, data processing, fault detection, communication with the cloud and monitoring via the Blynk interface. The first step in software design is including the necessary libraries like WiFi and Blynk libraries to enable wireless communication and connectivity with the cloud. Then the ESP32 is programmed with WiFi credentials and Blynk credentials. When the device successfully initializes, it connects to the cloud server using WiFi communication. For the sensor inputs, the ESP32 communicates with the LM35 temperature sensor and ACS712 current sensor. These two sensors are interfaced to analog input channels of the ESP32. In software, continuous reading of analog values of sensors is performed by the ESP32's built-in ADC function. The analog input values read are converted to voltage values and then to temperature and current values. The value of the temperature is calculated from the voltage reading obtained from the LM35 temperature sensor, which is directly proportional to temperature in degrees centigrade. Likewise, the value of the current is obtained from the current sensor sensitivity and offset voltage readings of the ACS712 current sensor. Techniques such as noise reduction and averaging are used in the software to increase sensor precision and stability in readings. The fault detection algorithm uses thresholds to classify abnormal operating conditions in the ESP32 program code. The software code continuously checks whether the temperature and current values exceed or fall within their corresponding thresholds. The result of this check enables the software to classify an operating state as NORMAL, WARNING, and CRITICAL Machine learning inference is not executed on the ESP32 during real-time operation. Instead, threshold-based logic is employed for deterministic fault detection, while machine learning analysis is performed to compare classification algorithms.

During normal operation, the ESP32 sends readings from the temperature and current sensors to the cloud server via the virtual pin interface at intervals. The Blynk application monitors the status of the sensor by displaying temperature, current, and status readings in its dashboard. When sensor readings surpass their threshold levels, the alarm buzzer is activated, and the dashboard updates the operating status. Similarly, the software implementation also implements edge-based processing whereby the information collected by the sensors is processed at the edge level by the ESP32 prior to uploading it to the cloud platform. This has the advantage of reducing delays and improving reliability. In conclusion, the software implementation can efficiently monitor, detect faults, connect with the cloud platform, and be remotely accessed.

B. Hardware Implementation

The Smart Vehicle Health Monitoring System consists of an ESP32 microcontroller, LM35 temperature sensor, ACS712 current sensor, BO motor, buzzer, power supply, and Blynk cloud platform. The ESP32 serves as the central processing unit, acquiring sensor data, performing threshold-based fault detection, and transmitting system status to the Blynk dashboard via WiFi. The LM35 sensor, connected to GPIO35, measures motor temperature, while the ACS712 sensor, connected to GPIO34, measures motor current to detect overload conditions. A BO motor is used as a laboratory-scale dynamic load to validate the sensing architecture under varying load conditions. A buzzer connected to GPIO2 provides immediate alerts during overheating or overload. Proper power supply and grounding ensure reliable operation and stable data acquisition. The hardware configuration enables low-cost, real-time vehicle health monitoring with edge-based fault detection and cloud-based remote visualization, making it suitable for predictive maintenance applications.

1) Overview of the System: The suggested Smart Vehicle Health Monitoring System is a smart IoT monitoring platform that aims to ensure continuous monitoring and detection of faults in the vehicles in real time. This IoT platform uses two sensors named LM35 and ACS712 for monitoring vehicle temperature and current respectively by interfacing with the ESP32 microcontroller board. The main aim of designing the suggested IoT-based system is to enhance the safety of the vehicles by detecting abnormal conditions like overheating and overloading of motors in the early stages itself. The ESP32 microcontroller gathers the data from the sensors and analyzes the collected data at the edge level and classifies the overall condition as normal, warning, or critical depending upon pre-defined threshold levels. This analyzed data from sensors is sent over WiFi to the Blynk platform which provides remote real-time monitoring and visualization of the system parameters. In case there is any abnormality like overheating and excess current, ESP32 triggers the alarm system to notify the user and changes the status on the dashboard. Therefore, it ensures timely identification of the fault and helps maintain the car system proactively. The suggested system involves a hierarchical Internet of Things structure that includes a sensing layer, edge computing layer, communication layer, cloud layer, and application layer. In the sensing layer, real-time data collection takes place, while in the edge layer, analysis and fault diagnosis occur locally. Data transmission in the communication layer takes place using WiFi technology, whereas remote monitoring via Blynk occurs at the cloud layer. Some benefits of this design include real-time monitoring, cost-effective construction, edge fault diagnosis, remote monitoring, scalability, and predictive maintenance. The use of IoT and embedded systems technology facilitates improved diagnostics of vehicles.

IV. Experimental Analysis

The proposed Smart Vehicle Health Monitoring System was evaluated using an ESP32, LM35 temperature sensor, ACS712 current sensor, BO motor, buzzer, and Blynk cloud platform. The BO motor was operated under different manually applied load conditions to validate the sensing architecture. Temperature and current were continuously measured using the LM35 and ACS712 sensors, respectively, while the ESP32 performed edge-based data processing and transmitted real-time sensor data to the Blynk dashboard via WiFi.

Under normal operating conditions, the system maintained low current and stable temperature, indicating a NORMAL state. As the motor load increased, the current rose, transitioning the system to the WARNING state. During overload or overheating conditions, the measured values exceeded predefined thresholds, resulting in a CRITICAL state and immediate activation of the buzzer. The Blynk dashboard continuously displayed temperature, current, and system status, enabling real-time remote monitoring. The experimental results demonstrate reliable sensing, low-latency cloud communication, accurate threshold-based fault detection, efficient edge processing, and timely alert generation, confirming the effectiveness of the proposed system for smart vehicle health monitoring and predictive maintenance.

V. Results and Discussion

A Smart Vehicle Health Monitoring System was developed using an ESP32, LM35 temperature sensor, ACS712 current sensor, BO motor, buzzer, and the Blynk cloud platform. The system continuously monitors vehicle temperature and current, performs threshold-based fault detection, and transmits real-time data to the Blynk dashboard via WiFi. The dashboard displays temperature, current, and system status with low communication latency.

Based on predefined threshold values, the operating condition is classified as **NORMAL**, **WARNING**, or

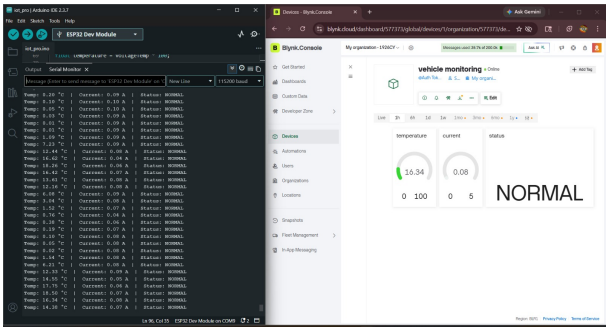


Fig. 3: Normal condition of vehicle

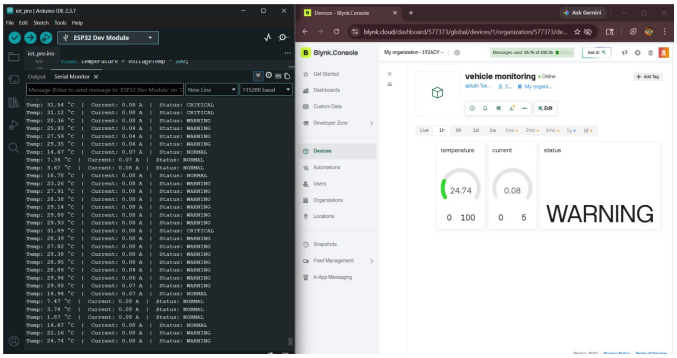


Fig. 4: Warning condition of vehicle

CRITICAL. Under normal conditions, the system operates with stable temperature and low current. As the motor load increases, the system transitions to the WARNING state, while overload or overheating conditions trigger the CRITICAL state, activate the buzzer, and update the cloud dashboard. The ESP32 performs edge-based fault analysis, reducing response time by processing data locally before cloud transmission. Experimental results demonstrate reliable real-time monitoring, accurate fault detection, and effective remote vehicle health monitoring through the Blynk platform.

1) Checking the output of the system: Output results of the proposed system were tested under different operating conditions by changing the motor load.

During the testing process, the LM35 sensor consistently provided real-time values of temperature readings, and the ACS712 sensor provided real-time values of current readings depending upon the load conditions of the motor. The processing of the obtained data was done by the ESP32 processor and was updated in the Blynk dashboard. Some observations that were observed during the testing process included: Fig.3. Under normal operational conditions, the status of the dashboard was NORMAL. Fig. 4.Moderate load conditions resulted in the WARNING status of the system. Fig.5. Overload and overheating conditions resulted in the CRITICAL status of the system along with activating the buzzer alarm system. Successfully sending of real-time sensor values to the Blynk dashboard using the WiFi module. Experimental output results proved the successful operation of the proposed IoT-based monitoring system.

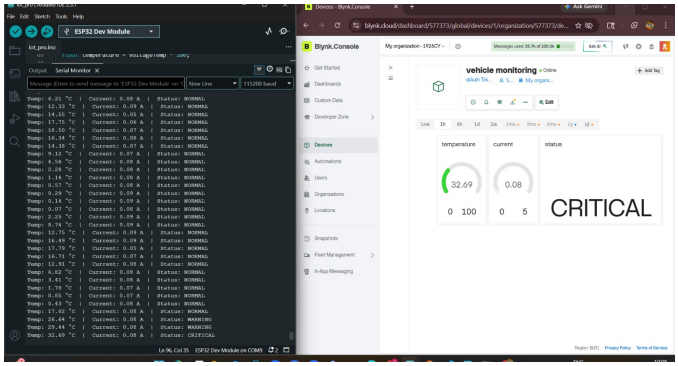


Fig. 5: Critical condition of vehicle

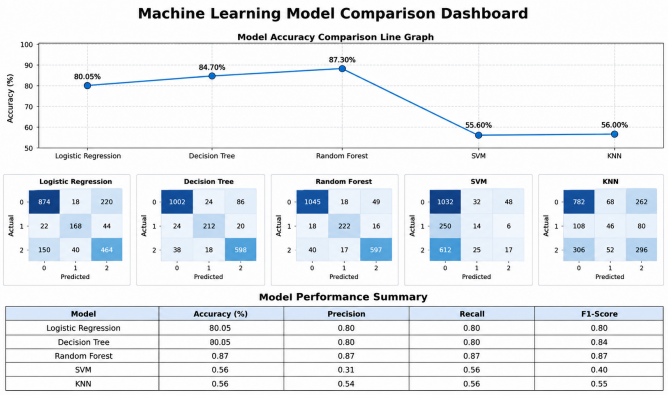
2) Efficiency of the System: The experimental results show that the designed system had very good efficiency in real-time monitoring and fault detection. The ESP32 processor worked very effectively in acquiring data from sensors, processing it, transmitting data through WiFi and updating the information on the cloud. These were successfully achieved: Real-time monitoring Low communication delay time Precise sensor data acquisition Effective communication with cloud Efficient classification of faults Generation of alerts in abnormal situations

3) Discussion: From the experimental findings, the Smart Vehicle Health Monitoring System proved to be an effective and reliable IoT-based framework for conducting real-time vehicle diagnostics and predictive maintenance. The use of both the LM35 and ACS712 sensors facilitated simultaneous monitoring of both temperature and current, leading to better fault detection accuracy. The ESP32 device was able to handle the tasks of edge-level computing as well as cloud communication efficiently without any delays. When compared to the conventional manual-based monitoring systems, the new design provided real-time monitoring capabilities, remote access, intelligent fault detection, and cloud-based visualizations. The implementation of Blynk helped to increase accessibility and remotely monitor vehicle operation status. The design was also scalable and flexible enough to facilitate the adoption in other future automotive applications. Other sensors as well as AI could be used to provide additional predictive maintenance features and smart transportation technologies.

A. Machine Learning Model Evaluation

Table I Accuracy Comparison of Machine Learning Models Table I compares the classification accuracy of five machine learning algorithms trained using the collected temperature and current sensor dataset. Among the evaluated models, Random Forest** achieved the highest accuracy of 87.3

Figure 6 presents a comprehensive comparison of the evaluated machine learning models using accuracy plots, confusion matrices, and performance metrics. Random Forest achieved the highest classification accuracy of 87.3%, with superior precision, recall, and F1-score compared to Logistic



Model Performance Summary

Model	Accuracy (%)	Precision	Recall	F1-Score
Logistic Regression	80.05	0.80	0.80	0.80
Decision Tree	80.05	0.80	0.80	0.84
Random Forest	0.87	0.87	0.87	0.87
SVM	0.56	0.31	0.56	0.40
KNN	0.56	0.54	0.56	0.55

Fig. 6: Machine Learning Model Comparison Dashboard

Regression, Decision Tree, Support Vector Machine, and K-Nearest Neighbor. The results demonstrate the effectiveness of Random Forest for comparative vehicle health classification using the collected sensor dataset.

TABLE I: Machine Learning Model Performance

Model	Accuracy (%)
Logistic Regression	78.5
Decision Tree	82.4
Random Forest	87.3
Support Vector Machine	84.1
K-Nearest Neighbor	80.6

VI. CONCLUSION

A smart Vehicle Health Monitoring System using IoT and Machine Learning has been effectively designed and developed using ESP32, LM35 sensor for temperature, ACS712 sensor for current, BO motor, buzzer alarming system, and Blynk Cloud Server. The Vehicle Health Monitoring System effectively monitored key parameters like temperature and current of the vehicle, detected faults at edge level, and delivered updates regarding vehicle health status in real time. ESP32 efficiently collected and processed the sensor data, classified the vehicle operational states as NORMAL, WARNING, and CRITICAL, and transferred real-time data to the cloud platform using WiFi communication. Buzzer alarm system effectively alerted the user when there was an overheating and overload state. To improve the ability to do predictive maintenance, machine learning was used in order to classify the vehicle health status. There were several machine learning methods including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and K-Nearest Neighbor that were tested on the obtained dataset. The most successful method among those mentioned above is Random Forest which showed the highest rate of classification accuracy of 87.3The designed system showed high results in real-time monitoring, visualization in the cloud, fault detection with low latency, and predictive maintenance in a cost-effective IoT platform. Thus, the integration of IoT and The present implementation performs real-time fault detection using embedded threshold logic. machine learning evaluation demonstrated the comparative performance of different classifiers and provides a foundation for future predictive maintenance systems using larger datasets and automotive-grade hardware. The future improvements can include AI analytics, vehicle tracking with GPS, vibration monitoring sensors, cloud data analysis, and deep learning algorithms for more precise fault prediction.

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